OPTICS · SEMESTER 5 · MODULE 2

Answer key: Interference

Formal solutions to all 42 questions of the model paper.

BEFORE YOU READ AN ANSWER

Each answer starts with **Know first**: topic codes from File 1. If one feels new, read it first, try the question yourself, then read the solution.

Grey mark notes show a typical split of marks (a guide, not an official scheme).

Section A · Short answers

2 marks each.

1 PYQ N19 Q15

Principle of superposition.

KNOW FIRST

T1 Superposition of two sinusoidal waves P1

Waves: phase and path

When two or more waves overlap at a point, the resultant displacement at that point at any instant is the algebraic (vector) sum of the displacements due to the individual waves: $\vec{y} = \vec{y}_1 + \vec{y}_2 + \cdots$. After crossing, each wave continues unaltered. Where the waves meet in phase the resultant is large (constructive interference); where they meet in opposite phase it is small or zero (destructive interference). This redistribution of light energy is interference.

N24 Q4 ("explain interference"): the same answer, with the definition: interference is the modification of intensity produced by the superposition of two or more coherent waves.

2 PYQ N25 Q2

Do intensities obey superposition?

KNOW FIRST

T1 Superposition of two sinusoidal waves T4 Intensity distribution P2 Intensity = amplitude squared, averaged

No. The principle applies to displacements (amplitudes, electric fields). Intensity is proportional to the square of the amplitude, and $(a_1 + a_2)^2 \neq a_1^2 + a_2^2$. For coherent waves $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta$, which can be up to $4I_0$ or as low as 0 for two equal waves. Intensities add simply ($I = I_1 + I_2$) only for incoherent waves, where the interference term averages to zero.

3 PYQ N25 Q3

Incoherent waves of equal intensity I.

KNOW FIRST

T2 Coherence T4 Intensity distribution

$I_R = I + I + 2I\langle\cos\phi\rangle$. For incoherent waves the phase difference φ changes randomly about 10^{10} times a second, so its time average $\langle\cos\phi\rangle = 0$. Hence the resultant intensity is $I_R = 2I$, uniform everywhere: no interference fringes.

4 PYQ N24 Q5

Coherence.

KNOW FIRST

T2 Coherence T5 Other fringe shapes; wide sources

Two waves are coherent if their phase difference remains constant in time. Light from an atom comes in wave trains of about 10^{-10} s, with random phase jumps between trains, so independent sources are incoherent; coherent sources are obtained by deriving two beams from a single source (division of wavefront or amplitude). **Temporal coherence**: the phase stays predictable along the beam for a coherence time τ_c, i.e. a coherence length $L = c\tau_c \approx \lambda^2/\Delta\lambda$. **Spatial coherence**: points across the beam are in step; it requires a narrow source.

5 PYQ N22 Q13

Conditions for sustained interference.

KNOW FIRST

T2 Coherence

1. The sources must be coherent (derived from a single source).
2. They must emit the same wavelength (monochromatic).
3. Amplitudes equal or nearly equal (good contrast).
4. Sources narrow and close together; screen far enough for visible fringe width ($\beta = \lambda D/d$).
5. Same state of polarisation, and path difference less than the coherence length.

6 PYQ N20 Q13

Division of wavefront and amplitude.

KNOW FIRST

T2 Coherence

Division of wavefront: the incident wavefront is divided into two parts by passing it through two narrow apertures, or by mirrors or prisms; the two parts, coming from different points of the same wavefront, are coherent and overlap (Young's double slit, Fresnel's biprism, Fresnel's two mirrors, Lloyd's mirror). **Division of amplitude**: the amplitude of the same wave is divided into two parts by partial reflection and partial transmission at a surface; the two beams travel different paths and recombine (thin films, Newton's rings, Michelson interferometer). Extended sources can be used with division of amplitude.

7

Coherence length.

KNOW FIRST

T2 Coherence

The coherence length is the length of an unbroken wave train (over which the phase is predictable): $L = c\tau_c$. Two beams split from one source interfere only if their path difference is less than L. For $\tau_c = 10^{-10}$ s: $L = 3 \times 10^8 \times 10^{-10} = 3 \times 10^{-2}$ m $= 3$ cm.

8 PYQ N25 Q14 + N22 Q7

Bandwidth; screen distance.

KNOW FIRST

T3 Path difference and fringe width M4 Units and powers of ten

$$\beta = \frac{\lambda D}{d} = \frac{500 \times 10^{-9} \times 1}{0.5 \times 10^{-2}} = 1 \times 10^{-4}\ \text{m} = 0.1\ \text{mm}.$$

False. β ∝ D: a screen close to the sources gives very narrow fringes that cannot be resolved, and the derivation also assumes D ≫ d. The screen must be far away.

9

Visibility.

KNOW FIRST

T4 Intensity distribution

$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2}$. With $I_1 = 4I_2$: $I_{\max} = (2+1)^2 I_2 = 9I_2$, $I_{\min} = (2-1)^2 I_2 = I_2$, so $V = \frac{8}{10} = 0.8$.

10

Shape of the fringes.

KNOW FIRST

T3 Path difference and fringe width

A fringe is the locus of points with constant path difference $S_2P - S_1P = \Delta$. Such points lie on a hyperboloid of revolution with S_1, S_2 as foci, so on a plane screen the fringes are hyperbolas, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (the central fringe $\Delta = 0$ is a straight line). Near the centre of a distant screen ($y \ll D$) the hyperbolas are almost straight lines parallel to the slits.

11

White-light fringes.

KNOW FIRST

T9 Interference with white light

At the centre the path difference is zero for every wavelength, so all colours are bright there and the central fringe is white. On either side a few coloured fringes appear, violet-edged inside and red outside, since each colour has its own fringe width $\lambda D/d$ (where Δ is half the violet wavelength violet is cancelled; where it is half the red wavelength, red). Further out many wavelengths are bright and many dark at each point, so the fringes overlap into uniform white. The white central fringe locates the zero path difference.

12

Fresnel's two mirrors.

KNOW FIRST

T7 Fresnel's two mirrors P5 Mirrors, prisms and lenses

The virtual sources S_1, S_2 are images of the same slit S in the two mirrors, so any phase change of S occurs in both at once: their phase difference is constant (coherent). If the mirrors are inclined at a small angle θ and S is at distance b from their line of intersection, S, S_1, S_2 lie on a circle of radius b and $\angle S_1 M S_2 = 2\theta$, so $d = S_1S_2 = 2b\theta$.

13

Dark fringe at the edge of Lloyd's mirror.

KNOW FIRST

T10 Lloyd's mirror T11 Phase change on reflection: Stokes relations

At the edge of the mirror the direct and reflected paths are equal, so a bright fringe is expected; it is observed to be dark. Hence the reflected wave has an additional phase change of π (path λ/2): light reflected from a denser medium (glass) suffers a phase change of π. All fringes of Lloyd's mirror are complementary to those of Young's arrangement.

14 PYQ N19 Q5 · adapted

Optical path from the lower surface.

KNOW FIRST

T12 A thin film at normal incidence P4 Refraction and optical path

Increases. The wave reflected from the lower surface travels twice through the film, an optical path $2nt\cos r$ ($2nt$ at normal incidence). Since optical path = n × geometric path, it increases with the refractive index n.

15 PYQ N19 Q13

Thin film at normal incidence.

KNOW FIRST

T12 A thin film at normal incidence T11 Phase change on reflection: Stokes relations

The ray reflected from the upper surface (air → film) suffers a phase change π; the ray reflected from the lower surface (film → air) does not, and has travelled an extra optical path 2nt. Effective path difference $2nt \pm \lambda/2$. Reflected light: **constructive** $2nt = (m + \frac{1}{2})\lambda$; **destructive** $2nt = m\lambda$, m = 0, 1, 2, ... (transmitted light: the reverse).

If $n_1 < n < n_3$ (film between a rarer and a denser medium), both reflections suffer π; the phase changes cancel and the conditions interchange: constructive $2nt = m\lambda$, destructive $2nt = (m + \frac{1}{2})\lambda$.

16 PYQ N22 Q27

Least thickness of the soap film for brightness.

KNOW FIRST

T12 A thin film at normal incidence M4 Units and powers of ten

Bright in reflection: $2nt = (m + \frac{1}{2})\lambda$. Least thickness, m = 0: $t = \frac{\lambda}{4n} = \frac{5839}{4 \times 1.42}$ Å.

ANSWER $t_{\min} = 1028$ **Å** $\approx 1.03 \times 10^{-7}$ **m**

17 PYQ N19 Q14

Colours of thin films.

KNOW FIRST

T16 Colours of thin films T13 Oblique incidence: the cosine law

When white light falls on a thin film (soap bubble, oil on water), light reflected from the upper and lower surfaces interferes. The path difference $2nt\cos r$ depends on the thickness t and the angle r. For a given t and r, wavelengths satisfying $2nt\cos r = (m + \frac{1}{2})\lambda$ are strongly reflected and those satisfying $2nt\cos r = m\lambda$ are absent, so the film looks coloured. Since t and r vary over the film, different places show different colours.

N22 Q8 ("due to dispersion", T/F): **False**, the colours are due to interference. N22 Q14, N24 Q7: same answer.

18 PYQ N19 Q17

No colours in thick plates.

KNOW FIRST

T16 Colours of thin films T2 Coherence

In a thick plate the path difference 2nt is many wavelengths. At any point, a large number of wavelengths spread over the visible spectrum satisfy the maximum condition and an equally large number the minimum condition; their mixture appears white. Moreover, the path difference exceeds the coherence length of ordinary white light, so the two reflected beams are not coherent. Hence no colours are seen.

19 PYQ N22 Q4 · adapted

Non-reflecting coating.

KNOW FIRST

T15 Non-reflecting (anti-reflection) films

Reflection (loss of light by reflection at the lens surfaces). The film thickness $\lambda/4n_f$ is chosen for the middle of the spectrum (yellow-green); for red and violet the condition is not met, so they are partly reflected, giving the coated surface a purple (red + violet) tint.

20 PYQ N20 Q10 · adapted

Fringes from two plane surfaces.

KNOW FIRST

T17 The wedge-shaped film

False. Two perfectly plane surfaces inclined at a small angle φ form a wedge whose thickness increases uniformly with distance. The fringes are straight, parallel to the edge and equally spaced, with width $\beta = \lambda/2n\phi$, independent of the order.

21 PYQ N19 Q3 · adapted

Central spot in Newton's rings.

KNOW FIRST

T18 Newton's rings T12 A thin film at normal incidence

At the point of contact the air film thickness is zero, so the geometric path difference is zero. In **reflected** light one of the two rays (reflected at the glass plate, air → glass) suffers a phase change π: the waves cancel and the centre is **dark**. In **transmitted** light there is no such relative phase change and the centre is **bright** (N20 Q3).

22

Ring spacing; liquid film.

KNOW FIRST

T18 Newton's rings

The radius of the m-th dark ring is $r_m = \sqrt{m\lambda R}$, proportional to √m. So $r_{m+1} - r_m$ decreases as m increases: the rings get closer outward (the film thickness $t = r^2/2R$ grows faster and faster). With a liquid of index n, $r_m^2 = m\lambda R/n$: every ring contracts by the factor $1/\sqrt{n}$, and $n = D_{\text{air}}^2/D_{\text{liquid}}^2$.

23 PYQ N25 Q4

Michelson: condition for constructive interference.

KNOW FIRST

T20 Michelson interferometer T13 Oblique incidence: the cosine law

$2d\cos\theta = (m + \frac{1}{2})\lambda$ for a bright fringe, where **d** = $x_1 - x_2$, the separation between mirror M_1 and the virtual image M_2' of M_2 (difference of the arm lengths); **θ** = angle of the ray with the normal to the mirrors (θ = 0 at the centre of the field); **m** = order of the fringe (an integer). The extra ½ arises from the π phase change of one beam at the beam splitter. (If that phase change is ignored, as in many books: $2d\cos\theta = m\lambda$ for a bright fringe.)

24

Compensating plate.

KNOW FIRST

T20 Michelson interferometer

The beam going to M_1 passes through the glass plate G_1 three times, the beam to M_2 only once. A plate G_2 of the same material and thickness, placed parallel to G_1 in the other arm, equalises the glass paths. For monochromatic light the difference could be made up by moving M_1, but since n varies with λ, this is impossible for all wavelengths at once. G_2 is therefore essential for white-light fringes.

Section B · Paragraph / problem
5 marks each.

25 PYQ N22 Q19

Superposition of two sinusoidal waves; conditions.

KNOW FIRST
T1 Superposition of two sinusoidal waves M1 Four trig identities M2 Waves as arrows (phasors)

Let $x_1 = a_1\cos(\omega t + \theta_1)$, $x_2 = a_2\cos(\omega t + \theta_2)$. By superposition, $x = x_1 + x_2$.

1. $x = (a_1\cos\theta_1 + a_2\cos\theta_2)\cos\omega t - (a_1\sin\theta_1 + a_2\sin\theta_2)\sin\omega t$
2. Put $a\cos\theta = a_1\cos\theta_1 + a_2\cos\theta_2$ and $a\sin\theta = a_1\sin\theta_1 + a_2\sin\theta_2$. Then $x = a\cos(\omega t + \theta)$: an SHM of the same frequency
3. Squaring and adding: $a^2 = a_1^2 + a_2^2 + 2a_1a_2\cos(\theta_1 - \theta_2)$; $\tan\theta = \dfrac{a_1\sin\theta_1 + a_2\sin\theta_2}{a_1\cos\theta_1 + a_2\cos\theta_2}$

Constructive $\theta_1 - \theta_2 = 2m\pi \Rightarrow a = a_1 + a_2$, $I \propto (a_1 + a_2)^2$ (path diff. mλ)

Destructive $\theta_1 - \theta_2 = (2m+1)\pi \Rightarrow a = |a_1 - a_2|$ (path diff. $(m + \frac{1}{2})\lambda$)

For $a_1 = a_2$ the destructive resultant is zero. The energy missing at minima appears at maxima: interference redistributes energy.

Marks: superposition set-up 1 · resultant amplitude and phase 2 · two conditions 2.

26

Fringe shift by a thin plate; mica thickness.

KNOW FIRST

T6 Shift of fringes by a thin plate T3 Path difference and fringe width P4 Refraction and optical path

1. Time from S_1 to P through the plate: $\frac{S_1P - t}{c} + \frac{t}{c/n} = \frac{S_1P + (n-1)t}{c}$, so the plate adds optical path $(n-1)t$
2. New central fringe O′ (zero path difference): $S_2O' - S_1O' = (n-1)t$
3. For a point at height x, $S_2P - S_1P = xd/D$ (Young). So $\frac{d}{D}\,OO' = (n-1)t \Rightarrow \Delta x = \frac{D(n-1)t}{d}$
4. Every fringe shifts by the same amount towards the covered slit; the fringe width is unchanged

Numerical: $t = \frac{d\,\Delta x}{D(n-1)} = \frac{0.1 \times 0.2}{50 \times 0.58} = \frac{0.02}{29}$ cm.

ANSWER $t = 6.9 \times 10^{-4}$ **cm** $= 6.9$ **μm**

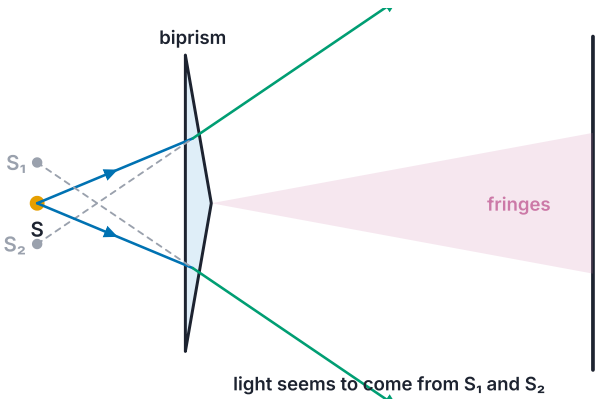
Marks: extra optical path 1 · shift formula 2 · numerical 2.

27 PYQ N20 Q19

Fresnel's biprism and wavelength.

KNOW FIRST

T8 Fresnel's biprism P5 Mirrors, prisms and lenses T3 Path difference and fringe width



Fresnel's biprism. Each half of the thin prism bends the light slightly. Behind it the light seems to come from two virtual sources S_1 and S_2 (grey dashed). They are copies of one source, so they are coherent. Fringes form where the two beams overlap (pink).

Principle. A biprism is two thin prisms of very small refracting angle α (about 30′) joined base to base. Light from a narrow slit S, parallel to the edge, is refracted by the two halves; the emerging beams appear to come from two virtual sources S_1, S_2 (division of wavefront). Being images of the same slit, they are coherent, and interference fringes are formed where the beams overlap.

Theory. Each half deviates light by $(n-1)\alpha$; with the slit at distance a from the biprism, $d = S_1S_2 = 2a(n-1)\alpha$. With the eyepiece at distance b beyond, $D = a + b$ and $\beta = \lambda D/d$.

Determination of λ. Slit, biprism and micrometer eyepiece are mounted on an optical bench and lit by sodium light; the slit is made parallel to the edge to get sharp fringes.

1. **β**: the cross-wire is moved across N fringes and β = distance/N.
2. **D**: slit-to-eyepiece distance from the bench scale.
3. **d**: a convex lens (f < D/4) is placed between biprism and eyepiece; at two positions it forms sharp images of S_1, S_2, with separations d_1 (magnified) and d_2 (diminished). Since the magnifications are v/u and u/v, $d = \sqrt{d_1 d_2}$.

ANSWER $\lambda = \dfrac{\beta\, d}{D} = \dfrac{\beta\sqrt{d_1 d_2}}{D}$

Marks: principle and diagram 2 · d and D 1 · measurement procedure and formula 2.

28 PYQ N19 Q26

Biprism fringe width.

KNOW FIRST

T8 Fresnel's biprism M3 Small quantities: three approximations M4 Units and powers of ten

Given n = 1.5, α = 1° = 0.01745 rad, a = 10 cm, b = 100 cm, λ = 5×10^{-5} cm

1. $d = 2a(n-1)\alpha = 2 \times 10 \times 0.5 \times 0.01745 = 0.1745$ cm
2. $D = a + b = 110$ cm

3 $\beta = \dfrac{\lambda D}{d} = \dfrac{5 \times 10^{-5} \times 110}{0.1745}$

ANSWER β = 0.0315 cm ≈ 0.32 mm

If "fringes at 1 m from the biprism" is read as D = 100 cm (slit distance neglected), β = 0.0287 cm. The source-to-screen distance a + b is the correct D.

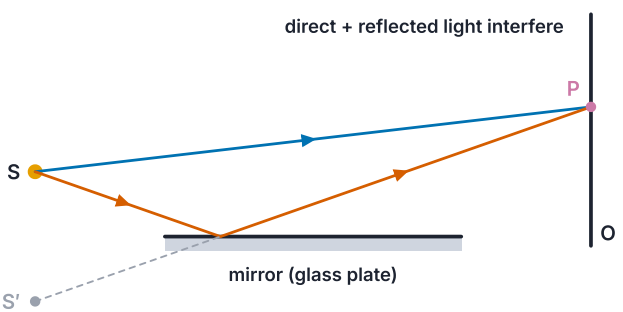
Marks: d 2 · D 1 · β 2.

29 PYQ N22 Q26

Lloyd's mirror fringe width.

KNOW FIRST

T10 Lloyd's mirror M4 Units and powers of ten



Lloyd's mirror. Light reaches P directly and after reflection at grazing incidence, as if from the virtual source S′. At O, where the screen touches the mirror's plane, the paths are equal, yet the fringe is **dark**: reflection added a phase change of π.

Arrangement. A slit S is held parallel to, and a small height h above, a plane glass plate (mirror). Light grazing the mirror is reflected and appears to come from the virtual image S′, a distance h below the mirror. S and S′ act as coherent sources $d = 2h$ apart; fringes form where the direct and

reflected beams overlap. Owing to the phase change π on reflection, the central fringe (at the mirror's edge) is dark.

1. $d = 2h = 4$ mm $= 4 \times 10^{-3}$ m, D = 1.5 m, λ = 589×10^{-9} m

2. $\beta = \dfrac{\lambda D}{d} = \dfrac{589 \times 10^{-9} \times 1.5}{4 \times 10^{-3}}$

ANSWER **β = 2.21×10^{-4} m ≈ 0.22 mm**

If the 2 mm is taken as the source separation d instead, β = 0.44 mm. The question gives the distance from the mirror, so d = 2h.

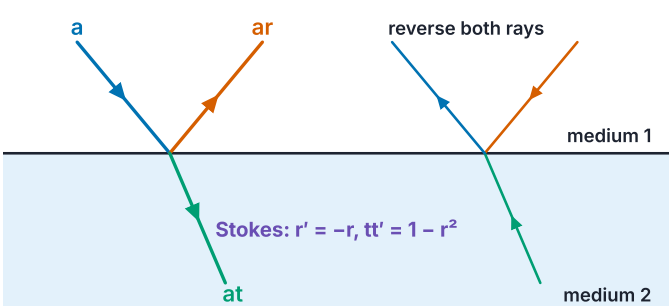
Marks: arrangement 2 · d = 2h 1 · β 2.

30

Stokes relations; phase change on reflection.

KNOW FIRST

T11 Phase change on reflection: Stokes relations



Stokes relations. Reverse the reflected and transmitted rays: they must rebuild the original ray only. This forces $r' = -r$: reflection from one side differs by a phase of π from reflection from the other side.

Let r, t be the amplitude reflection and transmission coefficients for light going from medium 1 to 2, and r′, t′ from 2 to 1. A wave of amplitude a gives reflected ar and transmitted at.

1. **Reversibility**: if the reflected and transmitted waves are reversed, they must recombine to give only the original wave reversed, of amplitude a
2. Reversed ar gives ar^2 (reflected, in medium 1) and art (transmitted, into medium 2). Reversed at gives att' (into medium 1) and atr' (reflected, in medium 2)
3. In medium 1: $ar^2 + att' = a \Longrightarrow tt' = 1 - r^2$
4. In medium 2 there must be no wave: $art + atr' = 0 \Longrightarrow r' = -r$

ANSWER $r' = -r,\ tt' = 1 - r^2$

Since $r' = -r = r\,e^{i\pi}$, the reflections at the two sides of an interface differ in phase by π. Lloyd's mirror (dark central fringe) shows that the phase change π occurs on reflection at the denser medium; reflection at the rarer medium gives no phase change. Consistent with Fresnel's normal-incidence result $r = (n_1 - n_2)/(n_1 + n_2)$, negative when $n_2 > n_1$.

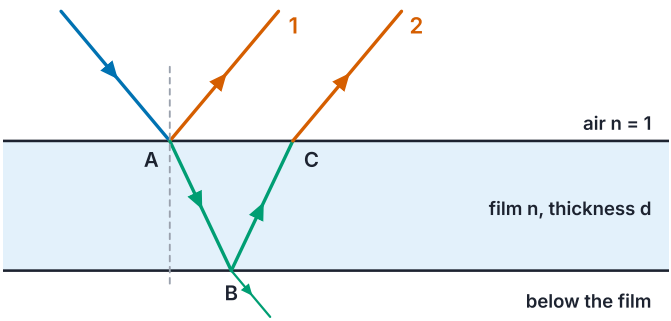
Marks: reversibility set-up 1 · two equations 2 · relations 1 · phase change 1.

31 PYQ N24 Q13

Plane parallel film, plane wave.

KNOW FIRST

T13 Oblique incidence: the cosine law T12 A thin film at normal incidence T2 Coherence



path difference of rays 1 and 2: Δ = 2 n d cos r

A thin film. Ray 1 reflects at the top surface; ray 2 goes into the film, reflects at the bottom and comes out. Ray 2 travels the extra optical path $2nd\cos r$ (cosine law), where r is the angle inside the film.

- A plane wave falls at angle i on a film of index n and thickness d. Each ray is divided at the upper surface: one part reflected (ray 1), one refracted, reflected at the lower surface and emerging parallel to ray 1 (ray 2). This is division of amplitude.
- The path difference is $2nd\cos r$ (cosine law), and ray 1 has an extra phase π (reflection at the denser medium). Reflected light: maxima $2nd\cos r = (m+\frac{1}{2})\lambda$, minima $2nd\cos r = m\lambda$; transmitted: the reverse.

- For a plane wave, i (hence r) is the same at every point; if d is uniform, the path difference is the same everywhere, and the whole film appears **uniformly** bright or dark. No fringes are formed; fringes appear only if the thickness varies (fringes of equal thickness).
- The eye sees only the light from a small region of the film which enters the pupil.
- If the film is thick, the path difference exceeds the coherence length and no interference is seen.

Marks: division of amplitude and diagram 2 · conditions 1 · uniform illumination and limits 2.

32

Two plane waves at ±θ.

KNOW FIRST

T5 Other fringe shapes; wide sources M1 Four trig identities T4 Intensity distribution

1. $E_1 = E_0\cos(kz\cos\theta + kx\sin\theta - \omega t),\ E_2 = E_0\cos(kz\cos\theta - kx\sin\theta - \omega t)$
2. Using $\cos A + \cos B = 2\cos\dfrac{A+B}{2}\cos\dfrac{A-B}{2}$: $E = 2E_0\cos(kx\sin\theta)\cos(kz\cos\theta - \omega t)$
3. Time average of $\cos^2(\ldots - \omega t)$ is ½: $I = 4I_0\cos^2(kx\sin\theta)$, with $I_0 = \dfrac{1}{2}KE_0^2$

4 Maxima where $kx\sin\theta = m\pi$: $x = \dfrac{m\lambda}{2\sin\theta}$; fringes are straight lines parallel to the z-axis, spacing $\beta = \dfrac{\lambda}{2\sin\theta}$

ANSWER θ = 30°, λ = 5000 Å: β = λ/(2 × 0.5) = 5000 Å = 0.5 μm

Marks: fields 1 · sum 2 · intensity and β 1 · numerical 1.

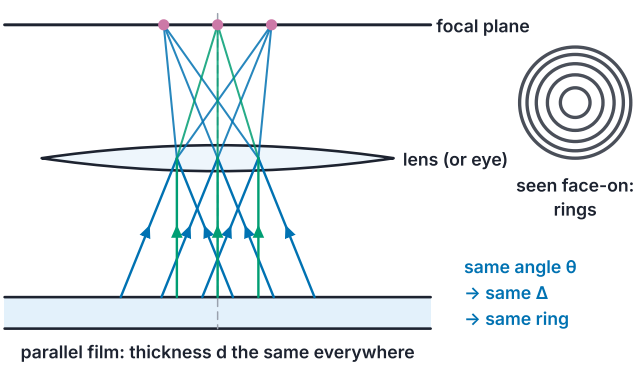
33 PYQ N24 Q16

Plane parallel film, point (or extended) source.

KNOW FIRST

T14 Point or extended source: Haidinger's fringes

T13 Oblique incidence: the cosine law



Haidinger's fringes (fringes of equal inclination). The film thickness is fixed; only the angle changes. All rays leaving the film at the same angle θ (blue, on either side) have the same path difference 2nd cos θ′ and are focused on one circle in the focal plane. So each fringe is a ring, seen focused at infinity.

- Light from a point source reaches the film at different angles. The reflected rays from the two surfaces have path difference $2nd\cos\theta'$, which now varies with the angle of refraction θ′.
- All rays making the same angle with the normal form a cone about the normal from the source, so the points of equal path difference lie on circles: the fringes are concentric **circular rings**. Dark rings: $2nd\cos\theta' = m\lambda$ (film in air).
- With an extended source, each point gives such rings, and they coincide only at infinity; the fringes are localised at infinity and are seen with the eye relaxed or a telescope focused at infinity. These are **Haidinger's fringes**, or fringes of **equal inclination**.
- The order is highest at the centre (θ′ = 0) and decreases outward.

N20 Q20 (conditions; difference from Newton's rings). Conditions: film of uniform thickness, broad monochromatic source, observation at infinity near normal incidence. Haidinger: constant thickness, fringes of equal inclination, localised at infinity, highest order at centre. Newton's rings: varying thickness, fringes of equal thickness, localised at the film, zero order (dark) at centre.

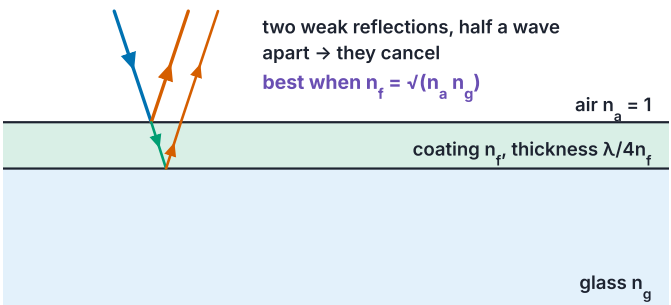
Marks: diagram 1 · path difference 1 · ring formation 2 · localisation 1.

34 PYQ N24 Q15

Non-reflecting films.

KNOW FIRST

T15 Non-reflecting (anti-reflection) films T11 Phase change on reflection: Stokes relations T12 A thin film at normal incidence



A non-reflecting (anti-reflection) film. Both reflections happen at a "rarer → denser" boundary, so both get the same phase change, which cancels out. The quarter-wave thickness makes them half a wave apart, so they cancel.

About 4% of light is reflected at each air–glass surface; in multi-lens instruments this loses light and produces flare. A thin transparent film of index n_f (between n_a and n_g, e.g. MgF_2, 1.38) on the glass reduces the reflection.

1. Both reflections (air → film, film → glass) are at denser media, so both suffer π: no net phase change

2. Destructive interference: $2n_f d = \frac{\lambda}{2} \Rightarrow d = \frac{\lambda}{4n_f}$ (quarter-wave film)

3. Complete cancellation requires equal amplitudes: $\frac{n_f - n_a}{n_f + n_a} = \frac{n_g - n_f}{n_g + n_f} \Rightarrow n_f = \sqrt{n_a n_g}$

For glass of 1.5, the ideal $n_f = 1.22$; MgF_2 (1.38) is used, with d ≈ 900 Å for λ = 5000 Å, reducing reflection from 4% to about 1.4%. The reflected energy is not destroyed but transmitted. Cancellation is exact for one wavelength (chosen in the yellow-green), so coated lenses look purple.

N19 Q22 (application in reducing reflectivity of lens surfaces): same answer.

Marks: purpose 1 · phase and thickness 2 · index condition 1 · remarks 1.

35 PYQ N20 Q31

Air wedge angle.

KNOW FIRST

T17 The wedge-shaped film M3 Small quantities: three approximations

1. At distance x from the edge, thickness $t = x\phi$. Dark fringes (reflected, normal incidence): $2t = m\lambda$, i.e. $2x_m\phi = m\lambda$

2. $2\phi(x_{m+1} - x_m) = \lambda \Rightarrow \beta = \frac{\lambda}{2\phi}$ (n = 1)

3. 100 fringes per cm ⇒ β = 0.01 cm. $\phi = \frac{\lambda}{2\beta} = \frac{6 \times 10^{-5}}{2 \times 0.01}$

ANSWER φ = 3×10^{-3} rad ≈ 0.17°

Marks: β formula 2 · substitution 2 · answer 1.

36 PYQ N22 Q28

Newton's rings: R and r_{30}.

KNOW FIRST

T18 Newton's rings P6 Geometry of a circle

1. Dark rings: $r_m^2 = m\lambda R$
2. $R = \dfrac{r_{20}^2}{20\lambda} = \dfrac{(1.1\times10^{-2})^2}{20\times670\times10^{-9}} = \dfrac{1.21\times10^{-4}}{1.34\times10^{-5}}$
3. $r_{30} = r_{20}\sqrt{\dfrac{30}{20}} = 1.1\times10^{-2}\times1.2247$

ANSWER R = 9.03 m; $r_{30} = 1.35\times10^{-2}$ m

N19 Q27 (D_{10} = 0.3 cm, λ = 6×10^{-7} m): $r_{10}^2 = 0.0225$ cm², $R = \dfrac{0.0225}{10\times6\times10^{-5}} = 37.5$ cm; film thickness $t = \dfrac{m\lambda}{2} = 3\times10^{-4}$ cm.

N20 Q26 (R = 100 cm, λ = 6×10^{-5} cm, bright rings $r^2 = (m+\frac{1}{2})\lambda R$, first bright ring m = 0): 9th bright, m = 8: $r = \sqrt{8.5\times6\times10^{-3}} = 0.226$ cm; 10th, m = 9: $r = \sqrt{9.5\times6\times10^{-3}} = 0.239$ cm. (Taking m = 9, 10: 0.239 cm and 0.251 cm.)

Marks: formula 1 · R 2 · r_{30} 2.

37

Newton's rings with the sodium doublet.

KNOW FIRST

T19 Two close wavelengths: rings fade and return

T18 Newton's rings

1. Rings of λ_1 = 5896 Å and λ_2 = 5890 Å become indistinct where the dark rings of one fall on the bright rings of the other: $\frac{2t}{\lambda_2} - \frac{2t}{\lambda_1} = \frac{1}{2}$

2. $t_0 = \frac{\lambda_1\lambda_2}{4(\lambda_1 - \lambda_2)} = \frac{5.896 \times 10^{-5} \times 5.890 \times 10^{-5}}{4 \times 6 \times 10^{-8}}$ cm $= 1.45 \times 10^{-2}$ cm

3. Ring number $m \approx \frac{2t_0}{\lambda} = \frac{2.89 \times 10^{-2}}{5.893 \times 10^{-5}} \approx 490$; radius $r = \sqrt{2Rt_0} = \sqrt{2 \times 100 \times 0.01447} = 1.70$ cm

ANSWER **t ≈ 0.145 mm, near the 490th ring, r ≈ 1.7 cm. The rings reappear at $2t_o$ ≈ 0.29 mm.**

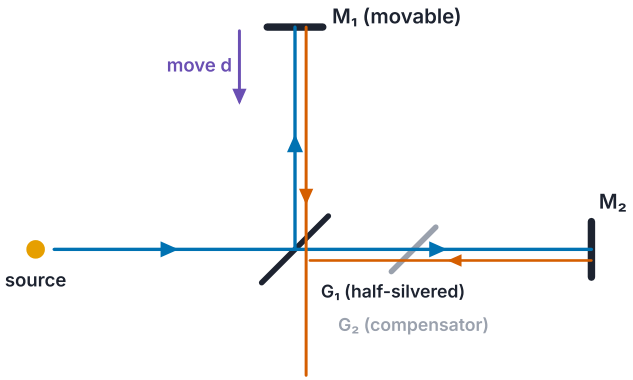
Marks: condition 2 · t_o 1 · m and r 2.

38 PYQ N22 Q20

Michelson interferometer; difference of wavelengths.

KNOW FIRST

T20 Michelson interferometer T21 Michelson: the difference of two wavelengths



Michelson interferometer. G_1 splits the beam into two (division of amplitude). After reflection at M_1 and M_2 the beams recombine and interfere. Moving M_1 by d changes the path by $2d$; each fringe that crosses the view means $2d = \lambda$ more.

Light from a broad source S is divided by the half-silvered plate G_1 (45°) into a reflected beam to M_1 (movable) and a transmitted beam to M_2 (fixed); G_2 is a compensating plate. The beams return to G_1 and recombine in the telescope. M_2' (image of M_2 in G_1) and M_1 form an air film of thickness d, giving fringes with $2d\cos\theta = m\lambda$ (dark).

1. Light with two close wavelengths $\lambda_1 > \lambda_2$ (sodium D lines). At d ≈ 0 (found with white light) both sets of rings coincide: distinct fringes

2. As M_1 moves, the fringes become indistinct when the bright rings of one fall on the dark rings of the other: $\frac{2d}{\lambda_2} - \frac{2d}{\lambda_1} = \frac{1}{2}$

3. Between two successive disappearances the mirror moves Δd, and the orders differ by one more: $\frac{2\Delta d}{\lambda_2} - \frac{2\Delta d}{\lambda_1} = 1$

4. $\Delta\lambda = \lambda_1 - \lambda_2 = \frac{\lambda_1\lambda_2}{2\Delta d} \approx \frac{\lambda^2}{2\Delta d}$

ANSWER $\Delta\lambda = \frac{\lambda^2}{2\Delta d}$; **e.g. Δd = 0.289 mm, λ = 5893 Å gives Δλ = 6.0 Å**

Marks: diagram and working 2 · disappearance condition 1 · derivation 1 · result 1.

Section C · Essays

10 marks each. About two pages in the exam.

39 PYQ N19 Q32 also N24 Q20

Coherent sources; resultant intensity; fringe width; incoherent sources.

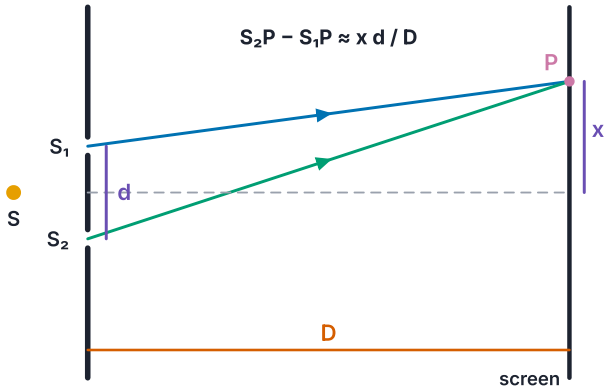
KNOW FIRST

T2 Coherence T3 Path difference and fringe width T4 Intensity distribution M1 Four trig identities P2 Intensity = amplitude squared, averaged

1. Coherent sources

Two sources are coherent if the phase difference between the waves they emit remains constant with time. Light from an ordinary source consists of wave trains about 10^{-10} s long with random phases, so two independent sources are incoherent. Coherent sources are produced from a single source by division of wavefront (Young's slits, biprism) or division of amplitude (thin films).

2. Path difference



Young's double-slit arrangement. Light from one slit S falls on two slits S_1, S_2 (distance d apart). They act as two coherent sources. On a screen at distance D, the point P at height x gets two waves with path difference $\Delta \approx xd/D$.

S_1, S_2 are separated by d; the screen is at distance D; P is at height x above the centre O.

1. $(S_2P)^2 - (S_1P)^2 = \left[D^2 + (x + \frac{d}{2})^2\right] - \left[D^2 + (x - \frac{d}{2})^2\right] = 2xd$
2. $S_2P - S_1P = \dfrac{2xd}{S_2P + S_1P} \approx \dfrac{xd}{D}$, since $S_1P + S_2P \approx 2D$ for $D \gg x, d$

3. Resultant intensity

1. $E_1 = E_{01}\cos\left(\frac{2\pi}{\lambda}S_1P - \omega t\right)$, $E_2 = E_{02}\cos\left(\frac{2\pi}{\lambda}S_2P - \omega t\right)$; by superposition $E = E_1 + E_2$
2. $I = K\langle E^2\rangle = K\left[\langle E_1^2\rangle + \langle E_2^2\rangle + 2\langle E_1E_2\rangle\right]$

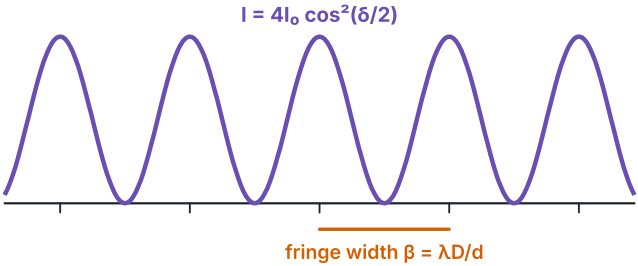
3. $2E_1E_2 = E_{01}E_{02}\left[\cos\delta + \cos\left(2\omega t - \frac{2\pi}{\lambda}(S_1P + S_2P)\right)\right]$, $\delta = \frac{2\pi}{\lambda}(S_2P - S_1P)$

4. Time averages: $\langle\cos^2\rangle = \frac{1}{2}$; the term in $2\omega t$ averages to zero

ANSWER $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta$, with $I_1 = \frac{1}{2}KE_{01}^2$, $I_2 = \frac{1}{2}KE_{02}^2$

4. Maxima, minima, fringe width

- Maxima: $\delta = 2m\pi$, i.e. $S_2P - S_1P = m\lambda$; $I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$.
- Minima: $\delta = (2m+1)\pi$, i.e. path difference $(m + \frac{1}{2})\lambda$; $I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$.
- Bright fringes at $x_m = \frac{m\lambda D}{d}$, dark at $\left(m + \frac{1}{2}\right)\frac{\lambda D}{d}$. Fringe width $\beta = x_{m+1} - x_m = \frac{\lambda D}{d}$, independent of m: the fringes are equally spaced.
- Equal intensities: $I = 2I_0(1 + \cos\delta) = 4I_0\cos^2\frac{\delta}{2}$, varying between $4I_0$ and 0.



Intensity on the screen. Bright fringes ($4I_0$) where $\delta = 0, 2\pi, \ldots$; dark (zero) where $\delta = \pi, 3\pi, \ldots$. Neighbouring bright fringes are β apart, all equally spaced.

The average intensity is $2I_0 = I_1 + I_2$: energy is redistributed, not created or destroyed.

5. Incoherent sources

If the sources are incoherent, δ varies randomly and rapidly, $\langle \cos\delta \rangle = 0$, and $I = I_1 + I_2$ (= $2I_0$ for equal sources), uniform everywhere: no fringes.

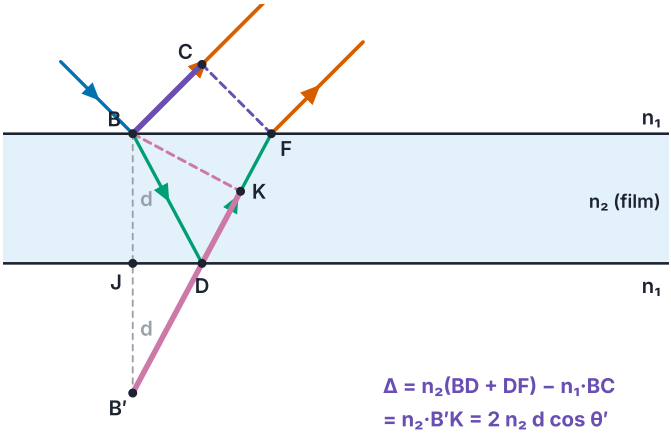
Marks: coherent sources 1 · path difference 2 · intensity derivation 4 · max/min, β 2 · incoherent 1.

40 PYQ N25 Q20 also N24 Q14

Maxima and minima for a parallel thin film; cosine law.

KNOW FIRST

T13 Oblique incidence: the cosine law T12 A thin film at normal incidence T11 Phase change on reflection: Stokes relations P4 Refraction and optical path



Cosine-law construction. One ray splits at B: reflected along BC (orange), refracted along B→D→F. FC ⊥ the reflected rays (outgoing wavefront). B′ is the mirror image of B in the lower surface (BJ = JB′ = d), so BD = B′D and B′, D, F lie on one line. BK ⊥ B′F. Snell's law makes $n_1 \cdot BC$ (purple, thick) = $n_2 \cdot KF$, which leaves the extra path $n_2 \cdot B'K$ (pink, thick).

A film of index n_2 and thickness d lies in a medium of index n_1. A ray incident at angle θ at B is partly reflected along BC and partly refracted (angle θ′) to D, reflected, and emerges at F parallel to BC. FC is drawn perpendicular to BC.

1. Path difference (cosine law)

1. $\Delta = n_2(BD + DF) - n_1\, BC$
   Beyond the wavefront FC both rays travel equal paths.

2. B′ is the image of B in the lower surface, so $BD = B'D$ and B', D, F are collinear: $BD + DF = B'F$

3. $\angle CFB = \theta$, so $BC = BF \sin\theta$. With BK ⊥ B′F, $KF = BF \sin\theta'$

4. Snell's law $n_1 \sin\theta = n_2 \sin\theta'$ gives $n_1 BC = n_2 KF$

5. $\Delta = n_2(B'F - KF) = n_2 B'K = n_2\, BB' \cos\theta' = 2n_2 d\cos\theta'$

ANSWER $\Delta = 2n_2 d\cos\theta'$ **(cosine law)**

2. Reflected light

For a film in air ($n_2 = n > n_1$), the ray reflected at B (rare → dense) suffers a phase change π; the ray reflected at D (dense → rare) does not (Stokes relations). The effective path difference is $2nd\cos\theta' - \lambda/2$.

Maxima $2nd\cos\theta' = (m + \frac{1}{2})\lambda$

Minima $2nd\cos\theta' = m\lambda$

3. Transmitted light

The two transmitted rays (one directly through, one after two internal reflections at D and F, both dense → rare) have the same path difference $2nd\cos\theta'$ but no relative phase change.

Maxima $2nd\cos\theta' = m\lambda$

Minima $2nd\cos\theta' = (m + \frac{1}{2})\lambda$

The reflected and transmitted systems are **complementary**: a thickness that is bright in reflection is dark in transmission, as required by conservation of energy.

4. Contrast

The two main reflected rays have nearly equal amplitudes (≈ 0.15a each for n = 1.36), so the reflected fringes have good contrast (minima nearly dark). The transmitted rays are very unequal (≈ 0.98a and 0.02a), so the transmitted fringes have poor contrast.

5. Remarks

- If the film lies between media with $n_1 < n_2 < n_3$, both reflections have π and the conditions interchange (used in non-reflecting films).
- Thick films give no fringes (path difference beyond the coherence length).

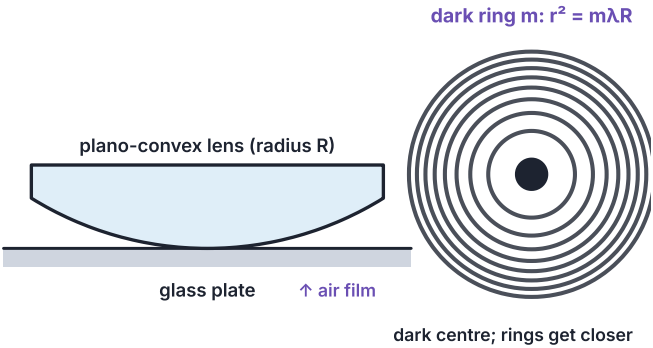
Marks: diagram 1 · cosine law 4 · reflected 2 · transmitted 2 · remarks 1.

41 PYQ N22 Q32

Newton's rings in reflected light; determination of λ.

KNOW FIRST

T18 Newton's rings T12 A thin film at normal incidence P6 Geometry of a circle

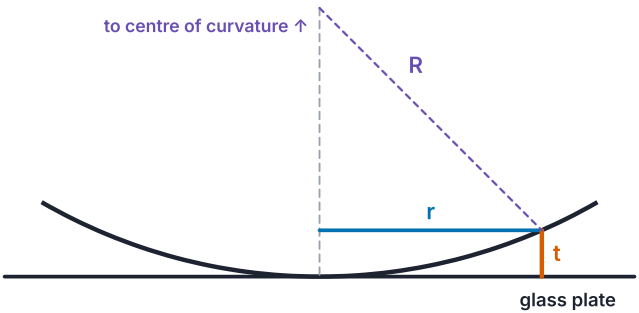


Newton's rings. A curved lens sits on a flat plate, trapping an air film whose thickness grows with distance from the centre. Equal thickness lies on circles, so the fringes are rings. The centre is dark in reflected light.

1. Arrangement

A plano-convex lens L of large radius of curvature R is placed, convex side down, on a plane glass plate P. Light from a sodium lamp is rendered parallel and reflected down onto the arrangement by a glass plate G at 45°. Light reflected from the top and bottom of the air film enters a travelling microscope above G. Concentric dark and bright rings are seen around the point of contact.

2. Theory



$R^2 = r^2 + (R - t)^2 \Rightarrow r^2 = t(2R - t) \approx 2Rt$

Newton's rings geometry. At distance r from the point of contact the air film has thickness t. Pythagoras in the big right triangle (sides r, $R - t$, hypotenuse R) gives $r^2 \approx 2Rt$.

1. At distance r from the point of contact the air film thickness t satisfies $R^2 = r^2 + (R - t)^2$, so $r^2 = 2Rt - t^2 \approx 2Rt$ (t ≪ R)

2. Normal incidence on an air film (n = 1): path difference 2t. Reflection at the lower surface (air → glass) adds π

3. Dark rings: $2t = m\lambda \Longrightarrow r_m^2 = m\lambda R$, $D_m^2 = 4m\lambda R$; so $r \propto \sqrt{m}$ and the rings crowd outward

4. Bright rings: $2t = (m + \frac{1}{2})\lambda \Longrightarrow r^2 = (m + \frac{1}{2})\lambda R$

t is constant on circles about the contact point ⟹ circular fringes (of equal thickness). At the centre t = 0 and only the π remains ⟹ dark spot.

3. Determination of λ

1. Focus the microscope on the rings. Set the cross-wire tangential to, say, the 20th dark ring on the left, and move across reading the positions of rings 20, 18, ..., 4 on the left and 4, ..., 20 on the right.
2. The diameter of each ring = difference of its left and right readings.
3. $D_{m+p}^2 - D_m^2 = 4p\lambda R$, so

ANSWER $\lambda = \dfrac{D_{m+p}^2 - D_m^2}{4pR}$

4. R is found with a spherometer. Alternatively, plot D_m^2 against m: the slope is $4\lambda R$.

Using the difference $D_{m+p}^2 - D_m^2$ removes the error due to imperfect contact (a dust particle or elastic deformation adds a constant to D^2) and the need to know the absolute order of a ring.

Extension: with a liquid of index n in the gap, $D_m^2 = 4m\lambda R/n$, so $n = D_{\text{air}}^2 / D_{\text{liquid}}^2$.

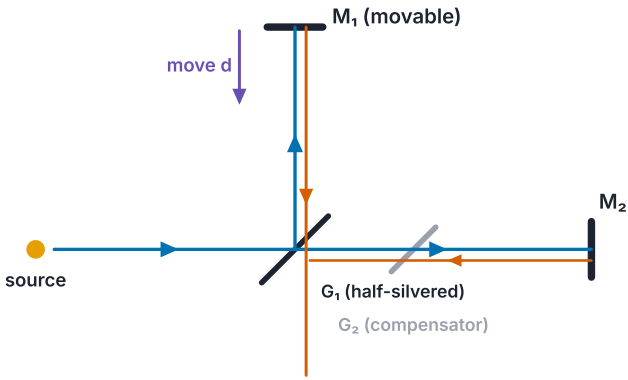
Marks: arrangement and diagram 2 · theory of radii 4 · centre dark, ring spacing 1 · determination of λ 3.

42 PYQ N20 Q32 also N19 Q21

Michelson interferometer; standardisation of the metre; circular, straight, white-light fringes.

KNOW FIRST

T20 Michelson interferometer T14 Point or extended source: Haidinger's fringes T17 The wedge-shaped film T9 Interference with white light



Michelson interferometer. G_1 splits the beam into two (division of amplitude). After reflection at M_1 and M_2 the beams recombine and interfere. Moving M_1 by d changes the path by $2d$; each fringe that crosses the view means $2d = \lambda$ more.

1. Construction and working

Light from a broad source S falls on G_1 (half-silvered on the back, at 45°), which divides it into a beam to mirror M_1 and a beam to mirror M_2 (division of amplitude). M_1, M_2 are perpendicular; M_1 moves on a micrometer screw, M_2 has tilt screws. The returning beams recombine at G_1 and interfere in the telescope T. G_2, identical to G_1, is the compensating plate.

2. Theory; measuring λ (N19 Q21)

The observer sees M_1 and the virtual image M_2' of M_2 formed by G_1. The arrangement is equivalent to an air film of thickness d between M_1 and M_2'. The path difference for a ray at angle θ is $2d\cos\theta$; including the π phase change at G_1, dark fringes satisfy $2d\cos\theta = m\lambda$.

1. At the centre (θ = 0): $2d = m\lambda$
2. M_1 is moved by d_0 and N fringes are counted crossing the centre: $2(d - d_0) = (m - N)\lambda$
3. Subtracting: $\lambda = \frac{2d_0}{N}$ (each fringe corresponds to a mirror movement of λ/2)

Example: N = 1000 for $d_0 = 0.290$ mm gives λ = 5800 Å.

3. Standardisation of the metre

Michelson and Benoit compared the standard metre with the red cadmium line (λ = 6438.4696 Å). Nine etalons (pairs of parallel mirrors), 10 cm, 5 cm, ... down to about 0.39 mm, were made. The wavelengths in the shortest were counted as fringes while M_1 moved from its front to its back mirror (white-light fringes marking each position). Each etalon was compared with the next longer one, and the 10 cm etalon was stepped ten times along the metre bar.

ANSWER **1 m = 1 553 164.13 wavelengths of the Cd red line (to about 1 part in 10^9)**

4. Types of fringes

- **Circular fringes**: M_1 parallel to M_2', monochromatic extended source. Parallel air film: the path difference $2d\cos\theta$ depends only on θ, so fringes of equal inclination, concentric rings seen at infinity. As d decreases, rings collapse into the centre and become broader; at d = 0 the field is uniform.

- **Straight fringes**: M_1 and M_2' slightly inclined, forming a thin air wedge, with d small. Fringes of equal thickness, straight and parallel to the line of intersection, localised near the mirrors. For larger d they become curved (convex towards the thin edge).
- **White-light fringes**: with white light, the wedge must have $d \approx 0$ (paths equal within a few wavelengths), since each wavelength has its own fringe width. A central achromatic (dark, because of the π phase change) fringe is seen, with a few coloured fringes on each side; beyond, the colours overlap into white. They locate the position of zero path difference exactly (used in the metre experiment). The compensating plate is essential for them.

Marks: diagram and construction 3 · theory 2 · standardising the metre 2 · three types of fringes 3.